

TECHNICAL NOTE · OPEN · V2

Hierarchical Bits — an investigation

From a representation paradigm to a property (FCIR), and its provisional conclusion

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An investigation reported at the honest size the evidence supports. It tests a strong hypothesis, narrows it under contrary evidence, names what survived (FCIR — a working name), and confronts it with the prior art that already implements it. Measured; with its boundaries and limitations stated.

Part I — The measured study

EXECUTIVE SUMMARY

Hierarchical Bits (BH) began as a hypothesis from December 2025 ("the byte is an implicit tree; hierarchy is interpretation"). This study tested the hypothesis from **nine independent angles**, with real measurements and honest baselines. The result is not "a better codec" — it is the precise delimitation of a paradigm:

BH does NOT compress dense signal better than JPEG/WebP/AVIF (it loses — measured).

BH IS a STRUCTURAL ENVELOPE that:

- (a) makes the structure explicit at low cost (0–6% of the file — measured),
- (b) routes each region's residual to the right specialist (orchestration),
- (c) offers multiple reads over the same structure (preview/ROI/proof).

Anchor result (measured): a structured document written in the orchestrated format is **2.1× smaller than WebP AND** lets you read any region in **3–55× fewer bytes**, within the same file — something no SOTA tool offers together.

The transversal law (measured across all angles): BH pays off exactly to the extent that the data is **structure** (rule-generated, composite, hierarchical) and not **signal** (noise/perceptual). At the limit, the payload becomes a *program* that generates the data (gains of thousands×); the boundary is not entropy — it is structure recognition.

Conceptual framing (added after this study). A later sweep of 20 domains (`applicability/`) reframed BH from a *file format* to a *property of representation* — the **First-Class Interpretation Representation (FCIR)**: rival interpretations kept as first-class, co-registered entities over one immutable substrate, without forced adjudication. The substrate-sharing economics this study measures are real but already mature SOTA in most domains; our sweep identified FCIR (a working name) as the property that best distinguishes BH from the domains evaluated — a result, not a universal claim. See `BH_PRINCIPLE.md` .

1. METHODOLOGY (what makes this study trustworthy)

The entire investigation followed four rules, applied without exception:

1.1 Declare before measuring. Each terrain defined falsifiable claims with a criterion for **success AND failure** BEFORE the measurement. No number was promised a priori.

1.2 Correctness as a gate. Before any performance comparison, exact correctness was verified (bit-for-bit reconstruction, equality of aggregates, valid cryptographic proof). Performance is only measured over a correct result.

1.3 Honest baseline, with the vs-naive / vs-SOTA distinction. Large numbers against naive baselines (full scan, transmitting everything) are marked as such and distinguished from gains against the state of the art (WebP, HNSW, OLAP).

1.4 Real measurement, public self-correction. Real hardware was used (RTX 3060, CUDA events, `nvidia-smi dmon`). When a measurement was biased in favor of the hypothesis, it was corrected in public. Three material self-corrections occurred and are documented (§7) — they are part of the evidence of rigor.

2. REFRAMING THE QUESTION

The initial question — **"is BH a better codec?"** — was answered, measurement after measurement, with **no**. The right question only emerged at the end of the investigation: **"is BH a structural orchestrator of representations?"** — and the answer is **yes**. The key was a separation that resolves the confusion of half the study:

RESIDUAL PROBLEM $\neq$ PARADIGM PROBLEM

Trying to compress the residual (dense signal) is the problem of specialist codecs — and BH loses at it. The BH paradigm is structure, belonging and reads. Separated, each half has an honest and opposite verdict.

3. THE TERRAINS TESTED (method + measurement)

3.1 Image codec — BH as a compressor (loses)

Method: quadtree codec with per-node interpretation selection (LEAF/RAMP/DCT), compared against PNG/JPEG/WebP on 4K frames, with matched PSNR.

Measurements: - Matched content (gradient, lossy): BH-ramp **2.4 KB @ 57.9 dB** vs WebP **116.7 KB @ 50.2 dB** → **48× smaller, with higher quality**. - Access (thumbnail of a natural 4K photo): BH reads **0.148 MB** vs WebP 0.664 (4×), JPEG 1.0 (7×), PNG 6.5 (44×). - Natural photo compression: **loses** (4–10× larger than WebP). - v0.3 (DCT interpretation): switching the error criterion from L_∞ to L_2 unlocked the DCT and cut 40–66%, but **did not close the gap** — residual entropy coding is missing. - Heterogeneous image (pure BH-compressor): **loses at all photo fractions (10.75× at 0% photo)** — because text/UI explodes the quadtree and WebP's entropy coding already makes the easy regions cheap.

Terrain conclusion: as a compressor, BH loses. The 48× gain on the gradient was a corner case (perfectly procedural content), not the rule.

3.2 Database — selective read by aggregation

Method: aggregation tree (min/max/sum/count per node) over 1,000,000 rows; metric = rows read; baseline = flat scan.

Measurements:

query	BH reads	flat reads	gain	vs
global SUM	0	1,000,000	∞	naive
SUM range 10%	1,564	99,868	64×	naive
COUNT key 2%	2,048	1,000,000	488×	naive
COUNT correlated value > p99	92,736	1,000,000	11×	naive
wrong axis / independent value / low selectivity	$\sim 10^6$	10^6	1–2×	boundary

Interpretation materialization: a per-region filter (not aggregated) read everything (1×); with a per-region counter in the node → root read, **0 rows (∞)** — at the cost of storage that scales with cardinality.

Honesty: the gains are enormous vs naive scan, but the mechanism (pre-computed summary) is already state of the art (zone-maps, OLAP, materialized views).

3.3 Verification / Merkle — selective proof

Method: Merkle tree (SHA-256, domain separation) over 1,048,576 items; crypto correctness gate (valid proof verifies, forged proof fails) before measuring; metric = bytes/nodes.

Measurements:

task	BH reads	naive baseline	gain
commitment of 1M items	32 B (1 hash)	33.5 MB	1,048,576×
membership proof	644 B (20 hashes)	33.5 MB	52,103×
locate tampering	40 nodes	1,048,576 re-hashes	26,214×

The proof grows with **log n** (1k → 10 hashes; 1M → 20). Honesty: it is the standard Merkle construction (blockchain/git/CT); the study shows the **unification** with the other terrains, it does not invent crypto.

3.4 Wafer — multiple co-registered layers + scale (film)

Method: quadtree-union over co-registered layers (RGB+depth+ segmentation); per-layer lossless reconstruction gate; measurement of shared-structure vs K independent trees.

Measurements: - Rigid union: co-registered **1.12×**; misaligned **0.32×** (**loses**); photo+lum+seg **0.67×** (**loses**). - With a derived layer (luminance from RGB): **0.78×**. - With base + local refinement (non-union): **1.12×** (**wins**) — the natural photo moves from loss to gain; verified by lossless

reconstruction. - **Scale proof (film, 720 frames = 30s):** independent $1.00\times$ · temporal (delta between frames) $1.65\times$ · wafer-still $1.12\times$ · **wafer+temporal $2.13\times$** . The structure-fraction rises from **16.7% → 33.4%** with the temporal — structure only starts to pay off when redundancy empties the payload.

3.5 GPU — data movement (read face, on real silicon)

Method: data-movement measurement (simulation) and real wall time (CUDA events, RTX 3060), with the load visible via `nvidia-smi dmon`.

Measurements: - Simulation (bytes moved, aggregation/LOD/context load): **$1,540\times$ less data**. - Light real test (1 GB in VRAM): total reduction flat $2,991\ \mu\text{s}$ vs BH $2.8\ \mu\text{s}$ → **$1,087\times$** ; range aggregation **$264\times$** ; flat bandwidth **$342\ \text{GB/s}$** (pinned at the ceiling of 360 → genuinely bandwidth-bound). - **Heavy test (real load):** 6,000 queries, **3.51 TB scanned**, SM at 100% for 26 s. Diagnosis of two pitfalls: (a) my flat kernel was suboptimal ($134\ \text{GB/s} = 37\%$ of the ceiling), inflating the raw number ($4,725\times$); (b) the BH side at the measurement floor (noise). **Honest triangulated number** (time vs flat-ideal AND data-moved: 3.51 TB vs $\sim 2\ \text{GB}$): **$\sim 1,750\times$** .

Extrapolation (published bandwidth, NOT measured): the batch gain is **hardware-invariant** ($\sim 1,755\times$ from the 3060 to the B200 — algorithmic, both scale with bandwidth); the single-query gain **shrinks** on fast cards ($992\times \rightarrow 45\times$), because the kernel-launch floor is fixed.

Native hardware model (labeled projection, not measurement): with the floor at memory-latency level ($0.3\ \mu\text{s}$, physical) instead of launch ($2.8\ \mu\text{s}$, software): single latency **$9,747\times$** , batch **$5,702\times$** (near-memory build), energy **$\sim 1,755\times$ less** ($\propto$ data moved — robust, independent of the pJ/B value).

Honesty: the large numbers are vs naive scan; the mechanism (pre-computed aggregate) is standard technique (OLAP/materialized views). The GPU proves that the **read face** is real and fast in hardware; it is not a standalone product.

3.6 AI multimodal storage — where BH was tested away from home

Method: unified substrate (structure + embeddings) vs stitched stack (storage + HNSW + cache + spatial index), over a co-registered asset.

Measurements (storage, slice 1):

d (embedding)	unified vs stitched
8	$4.0\times$ (wins)
128	tie
256	$0.86\times$ (loses)
768–4096 (real AI)	loses

Self-correction (progressive embedding / Matryoshka): by treating the embedding as compressible infrastructure (not irreducible payload) — prefix in the internal node instead of full-d —

the verdict **moves**: $d=256 \rightarrow 1.06\times$, $d=1024 \rightarrow 1.02\times$, $d=4096 \rightarrow 1.00\times$ (ties). The bulk (leaf embedding) is a Shannon tie; the gain is in the index.

Measurements (access, slice 2): preview $7.6\times$ · aggregate $3.2\times$ · ROI $2.1\times$ · narrow-scoped retrieval $1.4\times$ · **broad retrieval $0.23\times$ (loses)**.

Architectural debt decomposition (real RAG stack): the eliminable debt is **14–49%** (borderline), **dominated by embedding duplication** (dense payload), not by structure. Conclusion: dense AI storage is not BH's terrain.

3.7 Compositional / real density — the strongest signal and its limit

Method (compositional): symbolic data — concepts as equations of shared primitives — represented as an algebraic envelope vs dense vector.

Measurement: $384\times$ smaller (8 MB vs 3.07 GB for 1M concepts), and it answers structural queries ("which concepts use primitive X?") that the dense vector **cannot even formulate**.

Method (real density): intrinsic dimensionality of real embeddings (all-MiniLM-L6-v2 model, 6,000 PT words, PCA).

Measurement: 90% of variance in **160 of 384 dimensions** ($\sim 2.4\times$ compressible). **Honesty**: this is *linear* compressibility (already exploited by PCA/PQ/ Matryoshka), **not** symbolic compositionality. Real language data, in embedding geometry, is **not** compositional-symbolic — it is a subspace of medium dimension. The $384\times$ holds for data *that is* composition; how much of the real world is compositional remains the open bet (of the symbolic domain).

3.8 Decode-program — payload becomes program (the deep form of the law)

Method: the header carries the *program* that generates the region (not the result); the decoder executes the program. Test of generality (several families) + noise + hidden cost.

Measurements:

rule-generated family	WebP	program	gain
rings	37.8 KB	13 B	$2,904\times$
waves	46.4 KB	13 B	$3,572\times$
checkerboard	2.4 KB	3 B	$809\times$
gradient	2.1 KB	1 B	$2,096\times$

Under noise (procedural base + noise α): the program-aware approach **keeps the structure advantage** at all noise levels (subtracts the known base; whole-WebP pays for the structure it cannot separate). **Hidden cost (declared)**: the encoder needs to **discover** the program — the inverse problem, trivial for a known family, **undecidable in general** (Kolmogorov complexity is uncomputable). BH's boundary is not the residual's entropy — it is **structure recognition**.

3.9 Orchestration + union — the formulation that wins for the right reason

Method (orchestration): BH routes each region to the local specialist (flat→constant, gradient→formula, text→PNG, photo→WebP), compared against one-codec-on-the-whole, on two content types.

Measurements:

content	whole-WebP	orchestrated	result
photo-heavy	10.7 KB	10.8 KB	1.01× (ties)
document (closed forms)	4.8 KB	2.2 KB	2.1× smaller

Splitting into WebP per region (without routing paradigms) **loses** (1.10×) — the gain comes from cross-paradigm routing, not from the cut.

Method (envelope cost): decomposition of BH's bytes into framing / explicit structure / residual.

Measurement: explicit structure = **0–6% of the total** (3 images). The smart header is cheap; the expensive part was the residual (now delegated). *Answers the "killer question": making the structure explicit costs very little.*

Method (the union): document written as `.bh`, measuring the two faces in the same file.

Measurements: - Representation: **2.3 KB vs 4.8 KB for WebP = 2.1× smaller**. - Selective read: reading 1 region costs flat 87 B · gradient 96 B · text 590 B · diagram 1.8 KB — **3 to 55× less** than WebP, which needs the whole file for any region.

```
WebP/AVIF → optimal residual, SINGLE read
PDF       → orchestrates specialists, but flat list, no selective read
OLAP/GPU  → selective read, but not a representation format
.bh       → BOTH faces + hierarchy + belonging, in a single envelope
```

4. THE TRANSVERSAL LAW (measured across all angles)

Hierarchy pays off when STRUCTURE dominates the cost, not dense SIGNAL.

At the limit: payload → PROGRAM that generates the data, to the extent that the data is rule-generated. The boundary is not entropy — it is structure recognition.

data class	measured result
procedural / formula	tiny program: 800–3,600× (decode-program)
compositional / symbolic	envelope 384× smaller + queries the dense one cannot do
heterogeneous structured	orchestration 2.1× + selective read 3–55×
aggregable / reused	selective read ∞ –488× (database), 1,750× (GPU)
co-registered + redundant	wafer+temporal 2.13× (film)
dense / perceptual (photo, audio, embedding)	loses or ties — connectionism's territory

5. THE REORGANIZED THESIS (supported by the evidence)

BH is not a codec — it is a structural orchestrator. Two faces, measured:

- **Read face** (summary in the node → reading selectively is cheap): GPU 1,750×, database ∞ /488×, Merkle 52,000×, thumbnail/ROI 4–44×. Enormous numbers, but the mechanism is already SOTA.
- **Representation face** (explicit structure + delegated residual): orchestration 2.1× on a document. Smaller number, but **vs the state of the art**, by an architectural property.

The union of the two faces in one envelope (2.1× smaller AND navigable) is what no SOTA tool delivers. **The value is not in the compressed block — it is in the structure that knows what that block means.**

Three scales where BH survives scrutiny: 1. structural format (one asset) · 2. orchestrator (heterogeneous asset) · 3. compositional substrate (symbolic system). Common root: structure > signal.

6. HONEST BOUNDARIES (so the thesis does not turn into faith)

- "Wins for the right reason" ≠ "we found the product". It is scientific validation; the product requires construction and adoption (engineering, not benchmark).
- The orchestrator competes with PDF (the incumbent), not with WebP. Its differentiator — hierarchy + multiple reads — is real but unproven as a market need.
- The large numbers of the read face are vs NAIVE; vs SOTA, they tie. The differentiator is the UNION of the faces, not an isolated number.
- The decode-program requires RECOGNIZING the structure (inverse problem): trivial for a known family, undecidable in general.
- BH's home is STRUCTURE-DOMINANT data; in dense signal connectionism reigns, and BH DELEGATES to it — it does not confront it.
- A format dies without an ecosystem. The path is not a generic `.bh` competing with Parquet/PDF; it is building it native in a greenfield domain that already needs it.

7. THE SELF-CORRECTIONS (evidence of rigor, not of weakness)

Three measurements were corrected in public when they proved to be biased:

1. **GPU — suboptimal flat kernel.** The raw number (4,725×) was inflated by a poor baseline implementation; the honest triangulated number is ~1,750×.
 2. **Multimodal — embedding treated as irreducible payload.** Corrected with progressive representation (Matryoshka): storage moved from "loses" to "ties".
 3. **Heterogeneous — hypothesis "wins on mixed content".** Refuted by the number (loses 10.75×); the real gain requires per-specialist routing (orchestration), not pure BH-compressor.
-

8. RELATED WORK (situating vs the state of the art)

BH does not invent the techniques it uses — it **unifies** them. Each isolated capability already exists and is mature; the contribution is to bring them together in a single structural envelope. We situate here the four families of prior art.

Adaptive per-block codecs. JPEG (Wallace, 1991) established the model "transform + quantization + entropy coding" with a fixed global basis (DCT). Modern codecs — AV1/AOMedia (Chen et al., 2018), VVC, JPEG XL — do *mode decision* per block, choosing prediction/transform modes per region. The BH-codec competes directly here and **loses** (§3.1): it lacks the entropy-coding machinery, and per-region selection is what AV1 already does. The difference of BH is not compression — it is delegating the region to the specialist (§3.9).

Selective-read structures. The "read the summary, not the data" read is the basis of OLAP/data cubes (Gray et al., 1997), zone-maps and per-row-group statistics in Apache Parquet, and *materialized views*. The gains of §3.2 and §3.5 (database, GPU) are exactly this mechanism — enormous against naive scan, but **already state of the art**. We recognize them as a credibility anchor, not novelty.

Authenticated structures. The verification face (§3.3) is the Merkle tree (Merkle, 1987), the basis of Git, Certificate Transparency (Laurie, 2014) and blockchains (Nakamoto, 2008). We do not invent crypto; we demonstrate that the Merkle-proof is the *same* read-by-objective-over-hierarchy as the other terrains.

Compact representation and data-as-program. For embeddings, the compressibility that §3.6/§3.7 exploit is Product Quantization (Jégou et al., 2011) and Matryoshka Representation Learning (Kusupati et al., 2022); vector search is HNSW (Malkov & Yashunin, 2018). The "decode-program" of §3.8 is the direction of Kolmogorov complexity (Kolmogorov, 1965; Li & Vitányi) and has direct precedent in PostScript (page-as-program, Adobe) and procedural generation. Self-describing formats (HDF5, Protocol Buffers, PDF/ISO 32000) prove that "data that carries its own structure" works.

The gap that BH occupies. None of these unifies representation + selective read + specialist orchestration + hierarchy/belonging + proofs in a single envelope. PDF orchestrates but is a flat list

without selective read; OLAP navigates but is not a representation format; AV1 adapts per block but with a single family of modes. BH's contribution is the **union**, not the components.

9. REPRODUCTION

```
codec      X:\bitH\          py -m pytest tests -q      ·  src/bench/{harness,headtohead,portfolio,
                                     heterogeneous,cost_envelope,orchestrate,union}.py
database   X:\bitH\db\      py -m pytest tests -q      ·  src/bench/harness.py
merkle     X:\bitH\merkle\  py -m pytest tests -q      ·  src/bench/harness.py
wafer      X:\bitH\wafer\   py -m pytest tests -q      ·  src/bench/{harness,film}.py
gpu        X:\bitH\gpu\     py -m pytest tests -q      ·
src/bench/{real_gpu,heavy_gpu,extrapolate,native_model}.py
multimodal X:\bitH\multimodal\ src/{probe,fatia2,probe_progressive}.py
composit.  X:\bitH\compositional\
{probe_compositional,real_density,decode_program,program_test}.py
debt       X:\bitH\architecture_debt\ debt_decomposition.py

Python: X:/miniconda3/python.exe (NumPy, Pillow, hashlib, CuPy/CUDA, sentence-transformers,
sklearn)
Raw reports per terrain: RESULTS_*.md in each folder.
```

10. CONCLUSION

The study tested Hierarchical Bits from nine independent angles, with a declared method, correctness as a gate, honest baselines and public self-correction. The verdict:

BH is not a better codec — it is a different way to **organize and read structured data**. Where the data is structure (rule-generated, composite, hierarchical, aggregable, co-registered), the gain is real and sometimes by orders of magnitude, and its deepest form is the payload becoming the *program* that generates the data. Where the data is dense signal (photo, audio, embedding), connectionism reigns, and BH's role is to **delegate** to the specialist — not to compete. The unique contribution is not an isolated number; it is the **union**, in a single envelope, of lean representation, selective read, hierarchy and belonging — which no current tool offers together.

The final proof is not one more benchmark; it is the construction of the format in a domain where structure and belonging matter as much as size.

From thesis to artifacts — four usable prototypes

The construction is no longer one step but four, and they share **one envelope** — proof the paradigm generalizes across domains. Each is a Python library with a measured demo and tests; each reads only

the fraction a query asks for, in real bytes read from the file.

bhmem — **agent memory**. The agent writes structured memories (fact/event/relation + time + topic + provenance); the format exposes them through the multiple reads.

read	what it returns	bytes read	vs flat store (reads all)
<code>summary()</code>	summary of all topics	2.5%	35× less
<code>recall(topic)</code>	the memories of one branch	4.0%	22× less
<code>since(t)</code>	memories of the temporal window	9.8%	9× less
<code>provenance(id)</code>	source+path of 1 memory	10.8%	8× less

(Demo: ~90 days, 60 topics, 2,250 memories; 9/9 tests green.)

bhtrace — **distributed traces (observability)**. A trace is a tree (trace → spans → events); the structure is the index, the heavy attributes (SQL, stack traces, headers) are the residual.

read	what it returns	vs flat store
<code>summary()</code> / <code>critical_path()</code>	the skeleton without the payload	9× less
<code>subtree(span)</code>	a drill-in	9× less
<code>service(name)</code>	one service	5× less

(Demo: 269-span checkout trace, attribute-heavy; 9/9 tests green.)

bhckpt — **model checkpoints**. A checkpoint is a hierarchy (model → layers → tensors → experts); the weights dominate, the structure is tiny.

read	what it returns	vs flat checkpoint
<code>summary()</code>	architecture of a 16 MB model read in 9 KB, no weights	1,779× less
<code>expert(i, e)</code>	load one MoE expert without the mixture	20× less
<code>layer(i)</code> / <code>tensor(name)</code>	one layer / one tensor	12× / 10× less

(Demo: 4-layer transformer, 2 MoE layers × 6 experts, 16.2 MB; 8/8 tests green.)

bhanno — **adversarial annotations (the matrix of adverse interpretations)**. The strongest form of "multiple reads": the same substrate recorded **once**, with K annotators laying down co-registered layers that **disagree**. The wafer (§3.4) generalized from additive layers to rival ones.

read	what it returns	result
storage	K rival interpretations vs K independent copies	4.6× smaller (5 annotators)
<code>adjudicate()</code>	majority + agreement, reading labels only (no substrate)	11× less than full
<code>item_views(id)</code>	all K interpretations of one item (the matrix)	23× less

(Demo: 200 items, 6 KB substrate each, 5 annotators; 62% agreement, 75 contested; 9/9 tests green.)

Honest boundary, the same across all four: BH wins on structural access and **delegates** the dense residual (full-text search → inverted index; per-tensor quantization → its specialist; dense semantic recall → a vector index). For `bhckpt`, selective per-tensor read already exists (`safetensors`) — that is the *anchor*; the new piece is the *union* (hierarchy including the MoE expert as a first-class read, per-tensor codec routing, a Merkle face for provenance). For `bhanno`, the envelope *surfaces* the contest; it does not decide who is right.

These are not more measured angles — they are the thesis turning into tools, the **same envelope proven in four domains**. The gain scales with how structure-dominant the data is — the transversal law of §4. See the respective READMEs (`bhmem/` , `bhtrace/` , `bhckpt/` , `bhanno/`).

"The value is not in the compressed block. It is in the structure that knows what that block means."

Part II — The principle (FCIR)

First-Class Interpretation Representation (FCIR) — a working name

From a file format to a property of representation.

The applicability sweep (`applicability/`) established something uncomfortable and clarifying: the BH **shape** — a heavy substrate stored once, many co-registered layers, selective reads — is **already mature SOTA** in nearly every domain where it matters (DICOM, COG/STAC, lakeFS, CRAM/tabix, MAM, S-LoRA, named graphs). Selling BH as "substrate + overlays" is selling something the world already has.

So the contribution is not the format. It is a **principle** the prototypes are instances of — and principles survive where formats age.

The thesis

Hierarchical Bits is not a storage format. It is a way of representing data in which multiple concurrent — and possibly contradictory — interpretations share a single immutable substrate and persist as co-equal, first-class, individually addressable entities; adjudication between them is deferred to read-time and optional, never baked in and never destructive.

We give this property a **working name** — the **First-Class Interpretation Representation (FCIR)**: *a representation in which interpretations are first-class — persistent, addressable, co-equal — rather than temporary versions or conflicts to be resolved away.* "First-class" is the load-bearing word: it is what separates an interpretation with independent standing from one that exists only until it is merged into a single truth.

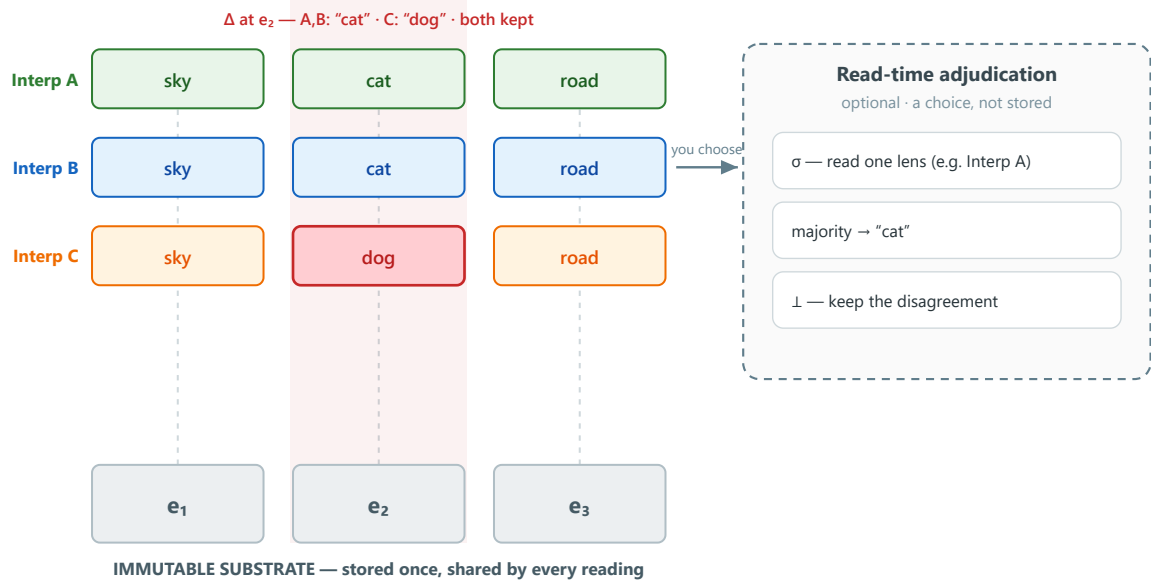
Why "working name." A property earns a permanent name after it has been observed, tested, and seen to recur across contexts — not at the moment it is first noticed. We are at the start of that sequence, not the end. So the honest statement is not *"BH's contribution is FCIR"* (a universal claim) but:

Our investigation identified FCIR as the property that best distinguishes BH from the approaches evaluated.

That is a description of a result, scoped to what we surveyed — not a law, and not a fence around what BH might later turn out to be. If the property proves broader or subtly different as it recurs in new contexts, the name should follow the idea, not constrain it.

First-Class Interpretation Representation (FCIR)

many interpretations · one immutable substrate · adjudication deferred and optional



$\oplus$ coexistence is decoupled from α adjudication — nothing is merged away; every reading persists and stays addressable.

The same model, as text (for readers — human or machine — that don't render SVG):

```
Interp A (alice): sky  cat  road   $\top$ 
Interp B (bob):   sky  cat  road   $\top$  coexist · co-registered · first-class
Interp C (carol): sky  DOG  road   $\perp$  (they disagree at  $e_2$  — both kept)
                |    |    |
substrate:        $e_1$    $e_2$    $e_3$     immutable · stored once
                |
read-time adjudication (OPTIONAL · not stored):
     $\sigma$  one lens    · majority → "cat"    ·  $\perp$  keep the disagreement
```

A worked example — 3 annotators, one image

```
Substrate: image #042 — stored once (the pixels)
Interp A (alice): { object: "cat" }
Interp B (bob):   { object: "cat" }
Interp C (carol): { object: "dog" }      ← disagreement, preserved

Reads (all over the one stored image):
layer("carol")      → { object: "dog" }
item_views("#042")  → { alice:"cat", bob:"cat", carol:"dog" } # the matrix
adjudicate(majority) → "cat"      # optional, at read time; the layers stay
disagreements()     → ["#042"]   # found by reading labels only, not the image
```

What FCIR changes vs today: current pipelines collapse A, B, C into a single gold label and **discard the fact that carol disagreed**. FCIR keeps all three first-class and lets each consumer adjudicate — or not — at read time. This is exactly the `bhanno` prototype.

The distinguishing test

What separates a BH representation from an `ANCHOR` system (one that *also* stores the substrate once and reads it selectively)?

Given two interpretations that **disagree** about the same element of the substrate, can **both remain** — permanently, queryable, neither marked wrong — until a reader chooses, or declines, to adjudicate?

- `ANCHOR` **systems answer no**. They converge to one canonical truth: a consensus mask, a gold label, a final coding decision, a merged model, a resolved clash. Disagreement is a *problem to be resolved* — noise on the way to ground truth.
- **BH answers yes**. Disagreement is *data to be preserved*. The rival readings coexist as permanent residents; the system never forces one to win by erasing another.

That single yes/no is the whole difference.

The four properties

#	property	shared with ANCHOR?
1	Immutable shared substrate — stored once, referenced by all layers	yes — already SOTA
2	Interpretations as first-class entities — persistent and addressable, not on-the-fly derived views nor mergeable annotations	partly
3	Coexistence of contradiction — rival readings of the <i>same</i> element are preserved together, as data	no — the differentiator
4	Deferred, optional adjudication — at read time the reader picks a lens: one interpretation / the matrix / the majority / the disagreement	no — the differentiator

Property 1 is the part that is already everywhere. Properties **3 and 4** are the part the sweep found **still under-explored**: most systems treat rival interpretations as a mess to clean up, not as co-equal entities to keep.

How FCIR relates to adjacent systems

A blind read test (five fresh reviewers, three model tiers, README only) converged on one verdict: the *message* is clear, but the *novelty* is not yet defended — every reviewer independently named existing

systems they believed already do this. They were right to. Here is the confrontation, under one rule: **where a system already does exactly this, we say so — no makeup.**

The FCIR property has five parts: (1) one shared, immutable substrate; (2) interpretations that may **contradict**; (3) kept as first-class, addressable, co-equal entities; (4) co-registered to the **same substrate elements**; (5) adjudication **deferred and optional** — neither reading marked wrong.

system	satisfies all five?	honest verdict
RDF named graphs + PROV	yes, for triples	already FCIR, in the triple/KG domain
Standoff annotation (W3C Web Annotation, UIMA/GATE)	yes, for text/media	already FCIR, in the annotation domain
Git / branches	no	defers adjudication, but is built for <i>eventual merge</i> ; branches are divergent whole-tree states, not co-equal readings co-registered per element
Bitemporal DBs / Datomic	partial	immutable + non-destructive history, but <i>temporal supersession of one truth</i> , not co-equal simultaneous rivals from different interpreters
CRDTs	no (opposite intent)	designed to auto-converge to one state without coordination; preserving disagreement is exactly what they remove (multi-value registers keep concurrent values only until the app resolves)
Event sourcing	no	immutable log + many derived projections, but the projections are different shapes of <i>one truth</i> , not contradictory rivals

The admission, without makeup. Two of these — **RDF named graphs with provenance** and **standoff annotation** — already implement the FCIR property in full, each within its own domain. Our `bhanno` prototype is standoff annotation; it did not invent it. For triple-shaped knowledge, named graphs got there twenty years ago. **So FCIR is not a new mechanism, and any claim that "nobody does this" is false.**

What is left, then — stated at its honest size. FCIR is an **architectural synthesis and a name**, not an invention:

1. a **single name** for a property that today exists only in domain-specific forms (named graphs for triples, standoff for text) and has *no shared term* across images, BIM models, model weights, agent memory;
2. a **falsifiable test** — *can two readings of the same element both remain, neither marked wrong, until adjudication is chosen?* — that classifies any system in one line;
3. the **observation** that the property recurs across domains that do not talk to each other, and that **most** systems in **most** domains do *not* have it (they converge, merge, or supersede).

That is a smaller claim than "a new representation paradigm." It may still be useful — naming a recurring property and giving it a test is how scattered practice becomes a discussable concept — but it

stands or falls as a synthesis, and an honest reader should judge it as one, not as a mechanism that did not exist before.

A **formal algebra** of these operators — coexistence $\oplus$, conflict Δ , projection σ , adjudication α , precedence $\triangleright$ — with FCIR stated precisely as $\oplus \perp \alpha$ (coexistence decoupled from adjudication), is in `BH_ALGEBRA.md`. It is what answers "*a model, or just a pattern?*" — as a specification, not yet a verified theory.

The prototypes are instances, not the point

Each `.bh` prototype is one instance of the principle — and they vary in how sharply they exercise properties 3–4:

instance	substrate	rival interpretations	adjudication deferred?
<code>bhanno</code>	a media item	K annotators that disagree	yes — <code>adjudicate()</code> is optional; the purest instance
<code>bhmem</code>	the interaction log	conflicting belief/fact versions over time	partial — rival temporal versions kept
<code>bhckpt</code>	base model weights	adapters / quantizations (alternative, mostly <i>additive</i>)	weak — interpretations rarely contradict
<code>bhtrace</code>	the span tree	competing analysis lenses / root-cause hypotheses	weak — lenses coexist more than contradict
<code>bhbim</code> (candidate)	the building object-graph	rival discipline/version overlays + clashes	yes — the one BUILD domain in the sweep

The closer an instance sits to "rival readings preserved without forced adjudication," the more it expresses what is *new* about BH versus the ANCHOR systems. `bhanno` (and the proposed `bhbim`) are the sharp instances; the others demonstrate the *generalization* but lean on the already-SOTA substrate-sharing.

Honest scope

- This is the differential **observed so far, in the 20 domains surveyed** — not a closed claim that properties 3–4 are BH's *only* possible value. New domains may surface other under-served slices.
- The substrate-sharing economics (property 1) remain real and measured in the prototypes — they are just **not novel**. BH's claim there is credibility by analogy, not invention.
- "Without forced adjudication" is a *representation* stance, not a truth claim: BH surfaces the contest and the majority; it does not decide who is right. Resolving truth stays a modeling choice the reader makes, or declines.

Why this is the advance

It moves the centre of gravity from the **file** to the **model**. If the thesis above can be stated precisely — and the distinguishing test makes it falsifiable: *does this system preserve contradicting interpretations without adjudication, or not?* — then the value of BH stops depending on any single prototype and starts depending on the clarity and generality of the principle. The prototypes become evidence; the principle becomes the contribution.

Companion to `BH_MASTER.md` (the measured study) and `applicability/` (the domain sweep). A Portuguese version can be produced on request.

Part III — The algebra of interpretations

Written in response to a fair external critique: *"without a formal algebra of how interpretations interact — composition, conflict, resolution, precedence — the proposal stays at the level of a design pattern, not a model."* This is the attempt to answer it. It is a **specification**, not a mechanized or proven theory (see *Honest limits*).

The whole point of FCIR is that adjudication is **deferred and optional**. An algebra makes that precise: it says exactly which operators a representation must provide, and which law makes it FCIR rather than something adjacent.

Objects

- **Substrate** S — an immutable set of addressable elements. A is its address space (ids, offsets, node paths, coordinates). S is never written.
- **Value/claim space** V — what an interpretation can assert about an address (a label, a value, a mask, a relation, a fact).
- **Interpretation** $I = (id, src, t, \pi)$ — a first-class entity with identity id , provenance src , time t , and a **partial** map $\pi : A \rightarrow V$ (defined only on the addresses it speaks about).
- **Interpretation set** $\mathcal{I}(S)$ — the interpretations currently co-registered to S . This set is the stored state.

Operators

1. **Coexistence** $\oplus$ (the primary operator) — $\mathcal{I} \oplus I = \mathcal{I} \cup \{I\}$. Non-destructive union. Adding an interpretation never alters or removes another. This is the operator most systems *lack* in non-destructive form.
2. **Restriction** $I|_{A'}$ — I narrowed to addresses $A' \subseteq A$.
3. **Conflict** Δ (a query, not a mutation) — for two interpretations, $\Delta(I_1, I_2) = \{ a \in \text{dom } \pi_1 \cap \text{dom } \pi_2 : \pi_1(a) \neq \pi_2(a) \}$; over a set, $\Delta(\mathcal{I}) = \bigcup_{i < j} \Delta(I_i, I_j)$. Detects disagreement **without resolving it**.
4. **Projection / lens** σ (read-only view) — $\sigma_\phi(\mathcal{I})$ reads S through the interpretations selected by filter ϕ (one id , one src , a time window). σ returns a view; it does not change $\mathcal{I}$.
5. **Adjudication** α_p (read-time, **optional**, policy-parameterized) — at an address a , $\alpha_p(\mathcal{I}, a)$ maps the multiset of claims $\{ \pi_i(a) \}$ to a single value **or** the explicit symbol $\perp$ ("undecided"). Policies p : `majority`, `priority(<)` (a precedence order over sources), `latest`, `trust(w)`, `unanimous` (returns $\perp$ iff there is conflict). α_p is a **function over the stored set**; it never mutates $\mathcal{I}$.

6. **Precedence / override** $\triangleright$ (derived view, non-commutative) — $I_1 \triangleright I_2$ is the reading in which I_1 wins where the two overlap. It is sugar for $\alpha_{\{\text{priority}(I_1 < I_2)\}}$; both interpretations remain in $\mathcal{I}$.
7. **Composition / stacking** $\circ$ — an interpretation may take **another interpretation as its substrate** (a review of annotator A's labels; a meta layer). $I_{\text{meta}} : \text{dom}(I_A) \rightarrow V$. This yields a finite stack; it is the least developed operator (see limits).
8. **Provenance** $\text{prov}(I) = (\text{src}, \text{t})$ — carried by every interpretation, preserved by $\oplus, \sigma, \alpha, \triangleright$.

Laws and invariants

- **(L1) Substrate immutability.** No operator writes to S .
- **(L2) $\oplus$ is a commutative, associative monoid** with identity $\emptyset$ (the empty interpretation), and idempotent on identical id (set semantics).
- **(L3) Non-destruction.** $\sigma, \alpha, \triangleright, \Delta$ are read-only: the stored set is invariant under them.
 $\text{view}(\mathcal{I}) \Rightarrow \mathcal{I}$ unchanged.
- **(L4) Conflict monotonicity.** $\Delta(\mathcal{I} \oplus I) \supseteq \Delta(\mathcal{I})$ — adding interpretations can only add conflicts, never silently erase them.
- **(L5) Adjudication separability.** α_p is definable as a pure function of $(\mathcal{I}, a, p)$. The stored set does **not** depend on p ; changing the policy changes only the view. This is the heart.

The FCIR test, made formal

A representation R is **FCIR** iff its coexistence $\oplus$ is non-destructive (L3) **and** its adjudication α is separable (L5) — i.e. $\oplus$ **and** α **are independent operators**.

Equivalently: *you can add a contradicting interpretation and still read every prior one unchanged, and you can change how you resolve conflicts without rewriting anything.*

This turns the earlier prose test ("can two readings of the same element both remain until a reader chooses to adjudicate?") into an algebraic property, and it **explains the prior-art verdicts as facts about coupling**:

system	$\oplus$ and α	FCIR?
RDF named graphs + PROV	decoupled: $\oplus$ = add a graph; α = query-time choice	yes
Standoff annotation	decoupled: $\oplus$ = add a layer; α = optional gold/consensus pass	yes
Git / branches	coupled: a branch defers α , but the model drives toward <code>merge</code> (= α at integration time); unmerged = divergent copies	no
CRDTs	coupled: α is baked into $\oplus$ (deterministic auto-convergence)	no
Event sourcing	single truth: projections are σ over one history; no rival α	no
Bitemporal / Datomic	coupled to time: α = temporal order; supersession, not co-equal rivals	partial

So FCIR is not "we invented coexistence." It is: **the property is exactly the decoupling of $\oplus$ from α , and that decoupling already exists in named graphs and standoff annotation — but is absent from the systems people reach for by default** (Git, CRDTs, event sourcing, bitemporal stores).

What the algebra buys

- It moves FCIR from *pattern* to *model*: a named set of operators with laws and a single defining property ($L_3 \wedge L_5$).
- It makes the test **mechanically checkable**: implement $\oplus$, Δ , α_p ; verify L_3 (views don't mutate) and L_5 (policy is a read-time parameter).
- It **predicts**, rather than asserts, which systems qualify — and why.

Honest limits

- This is a **specification, not a verified theory**. There are no consistency, soundness, or complexity results here, and no mechanized proofs.
- **Composition (◦) is under-specified** — interpretation-of-interpretation needs a typing discipline to avoid paradox; only the finite-stack case is sketched.
- **No implementation realizes the full algebra**. `bhanno` implements a fragment: $\oplus$, Δ , and α_{majority} / $\alpha_{\text{unanimous}}$. The other prototypes use even less of it.
- The algebra **does not establish novelty**. Named graphs and standoff annotation satisfy $L_3 \wedge L_5$ already; the algebra's contribution is to *state the property precisely and cross-domain*, not to claim it is new.

In short: this answers "*is there a model, or just a pattern?*" with **"here is a model, written as a specification"** — while being explicit that a specification is not yet a proven theory, and that the property it formalizes is, in two domains, already implemented.

Part IV — Provisional conclusion

This project began as a strong hypothesis and is reported here at the honest size the evidence supports — no larger.

Initial hypothesis. BH introduces a new, general paradigm of representation.

Result of the investigation. That hypothesis was **not confirmed**. The principles it leaned on — a shared substrate stored once, and selective reading over it — already appear, in mature form, across many domains (DICOM, COG/STAC, lakeFS, CRAM/tabix, S-LoRA, MAM, ...). As a *general paradigm*, BH is not novel.

What remained. A recurring property around the representation of **concurrent, possibly contradictory interpretations** over one shared substrate. It appears in some specific domains — notably **RDF named graphs (+ provenance)** and **standoff annotation** — but has no uniform, named treatment across them. We give it a working name: the *First-Class Interpretation Representation (FCIR)*.

Current status. There is **not yet sufficient evidence** that this property constitutes a *new* concept. The most defensible contribution at this moment is to treat it as a proposal of **synthesis and classification** — a cross-domain name plus a falsifiable test — subject to further validation. Where existing systems already implement it, this report says so (see [BH_PRINCIPLE.md](#)).

The conclusion, in three lines

- BH as a **universal paradigm of representation** was **not confirmed**;
- BH surfaced a **recurring representation property** worth study, whose **novelty is not established**;
- the work remains valid **as investigation and as method**, independent of whether the property turns out to be novel.

This is a strong conclusion precisely because it is compatible with everything observed. It does not overstate what was found, and it does not discard the work done.

What the research is, regardless

A hypothesis was pursued, pressure-tested, and cut down to what survived: hypotheses tested, counter-examples sought, scope reduced when the evidence demanded it, contrary evidence incorporated rather than explained away. The honest win is **the method** as much as any single finding — and a measured trail with its boundaries and its prior art named in public.

Responding to external critique (DeepSeek review, 2026-06)

An external model-based review raised five points. Two were already addressed in this repo (immutability is not new → the prior-art confrontation in `BH_PRINCIPLE.md`; the 20-domain method → `applicability/`). Three were fair and are addressed here:

(a) "No formal algebra → a pattern, not a model." Addressed in `BH_ALGEBRA.md`: the operators ($\oplus$ coexistence, Δ conflict, σ projection, α adjudication, $\triangleright$ precedence, $\circ$ composition), the laws, and **FCIR stated formally as the decoupling of coexistence from adjudication** ($\oplus \perp \alpha$). It is a specification, not yet a verified theory.

(b) Trade-offs (the costs, honestly). The family is not free: immutability → monotonic storage growth, needing compaction/GC and content-addressing; co-registration → interpretations must address substrate elements stably, so substrate re-addressing breaks layers (the standoff "offset-drift" problem); preserving conflict → read-time adjudication is $O(\text{layers})$ per address. None of these are solved here — they are the known costs of the immutable-substrate family.

(c) Comparison to real tools, not naive baselines. The prototype numbers are vs a *naive flat store*, not vs SOTA. Said plainly: **against the real tools the prototypes do not win.** `bhanno` is standoff annotation (brat / Label Studio / Web Annotation do it better and at scale); `bhckpt`'s selective load is what `safetensors` / S-LoRA already ship; `bhmem` sits below Memo / Zep. The prototypes are **illustrations of the property, not competitors**; the claim is the property + the test, not the implementations.

(d) Where it would actually help. The most defensible use case is **preserving and learning from disagreement** — *human label variation* in ML. Pipelines today collapse many annotators into one gold label and discard the disagreement, which a growing literature argues is real signal (ambiguity, subjectivity, uncertainty) being thrown away. FCIR's narrow but concrete value: keep every annotator's reading first-class and adjudicate **at read-time, per consumer** (one model trains on majority, another on the full distribution, an auditor inspects the conflict) — without forking the dataset. The usual gap is not storage (standoff already stores layers) but the *pipeline* collapsing them to gold. That gap, not the bytes, is the opportunity.

This does not make FCIR novel (see the prior-art confrontation). It makes the critique part of the record.

Paths from here (open)

None of these requires the original hypothesis to have been right:

- **Conclude** — report the investigation as-is; a finished inquiry.
 - **Reposition** — treat BH as a *lens for studying representation properties*, not a format or an invention.
 - **Use it** — take the parts that work and put them in software; not every concept needs to become a paper.
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Companion to `BH_MASTER.md` (the measured study), `applicability/` (the domain sweep) and `BH_PRINCIPLE.md` (the FCIR confrontation). Provisional — it should be revised if stronger evidence, for or against, appears.